CASE STUDY

How a Data Cleaning Mistake Can "Invent" a Patient.

And how I cleaned 1,000 messy Electronic Health Records (EHR) in Python without fabricating data.

98.3%
COHORT RETAINED

1,325 → 0
NULLS RESOLVED

100%
AUDIT PROVENANCE

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SWIPE ➡️

01 • THE PROBLEM

The Messy Reality of Raw EHRs

Real healthcare data is full of traps that break analytical models.

Composite Blood Pressure
Stored as raw unstandardized slash string; unusable for statistical math.
`"120/80"`

Verbal Words in Numeric Columns
Text values mixed with float decimals and null entries.
`"forty"`

Polymorphic Timestamps
Inconsistent delimiters and underscore formatting across clinics.
`"2023_04_15"`

THE CORE DANGER:
Blindly calling `df.fillna()` assigns artificial clinical profiles to blank records. You end up "inventing" fake patients.

02 • CLINICAL TRIAGE

Pruning the "Ghost Patients"

How I established a medical triage filter instead of blind deletion.

17

"Ghost Records" Safely Pruned (1.70%)

983 PATIENTS KEPT

Preserved 98.3% of the real patient cohort.

THE OBJECTIVE TRIAGE RULE:

If a patient is missing ≥ **3 core clinical markers** (`Age`, `Blood Pressure`, `Condition`, `Cholesterol`):

➔ **Prune the record.** They have no clinical substance to impute.

✓ Naive deletion would have dropped ~10% (99 rows). Our triage only pruned 1.70%.

03 • METHODOLOGY

Hierarchical Cascading Imputation

Why universal mean imputation is dangerous in healthcare.

A 25-year-old female and a 75-year-old male have completely different healthy vital baselines. Flat averages destroy this distinction.

TIER 1

(Age Group + Biological Sex) Subgroup Median

Most accurate clinical estimate based on demographic peer cohort.

TIER 2

Biological Sex Fallback Median

Applied only if age subgroup data is sparse.

TIER 3

Global Population Median

Final safety fallback ensuring zero unresolved values.

04 • MEASURABLE IMPACT

Measurable Transformation

From raw, unformatted records to analytics-ready clinical data.

1,325 → 0

MISSING VALUES
100% resolved across all 15 clinical attributes.

98.3%

COHORT PRESERVED
983 clean records ready for predictive modeling.

100%

VITALS USABILITY
Extracted systolic & diastolic numeric features.

HIPAA

AUDIT PROVENANCE
All imputed cells tracked via explicit Yes/No flags.

05 • REPOSITORY & CODE

Key Takeaways & GitHub

What this project proves about production data engineering.

1. Domain Rules Beat Generic Imputation

Healthcare data requires medical logic. Pruning 1.70% of unrecoverable rows protected data integrity without sacrificing sample size.

2. Explainability is Mandatory for Regulated AI

Stratified medians with explicit `*_imputed` audit flags ensure complete data provenance required by HIPAA and FDA 21 CFR Part 11.

OPEN-SOURCE CODE & NOTEBOOK:

github.com/ELSHINE123/healthcare-data-cleaning-pipeline

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